MTH 335(A) Dr. James Ashe

Practice Midterm March 22, 2013 **Name:**

1. Determine the following

a. 17 mod 2

b. $\mathrm{lcm}(7^{4000} \cdot 5^2 \cdot 2^4, 7^4 \cdot 5^{1000} \cdot 2)$

c. $17 = x \cdot 2 - y \cdot 15$. What is $\gcd(x, y)$?

2. Complete the following

a. Write the given set in tabular form: $A = \{x \mid x \in \mathbb{N} \text{ and } x \leq 5\}$

b. Write the given set in set-builder form: $A = \{\cdots - 2, 0, 2, 4\}$

3. Let $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$, $C = \{4, 5, 6\}$.

a. Find $A \cup C$

b. Find $A \cap B$

c. Find $A \cap C$

d. Find $(A \cup C) \cap B$

4. Let A and B be two sets. Prove that $A \cap B \subseteq A$.

5. Let $A \neq \emptyset$ be a subset of B. Prove that $A \cup B = A$.

6. Give the definition of the following:

a. A *function*

b. An *onto function*.

7. Consider the map $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined as $f(x, y) = xy$. Then f is:

(a) an onto function.

(b) a 1-1 function.

(c) a bijection.

(d) none of the above.

8. Let $f : A \to B$ be a function, $A = \{1, 2, 3\}$, and $B = \{1, 2\}$.

a. Can f be 1-1? Explain.

b. Show that f can be onto.

9. Define the following relation on the real number set: $x\mathcal{R}y \iff xy > 0$. Show the following:

a. $\mathcal{R}$ is *not* reflexive.

b. $\mathcal{R}$ is transitive.

c. $\mathcal{R}$ is *not* an equivalence relation.

10. Give an example of a relation on a set which is an equivalence relation.